ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C4 · CIRCUITS AND MAGNETS

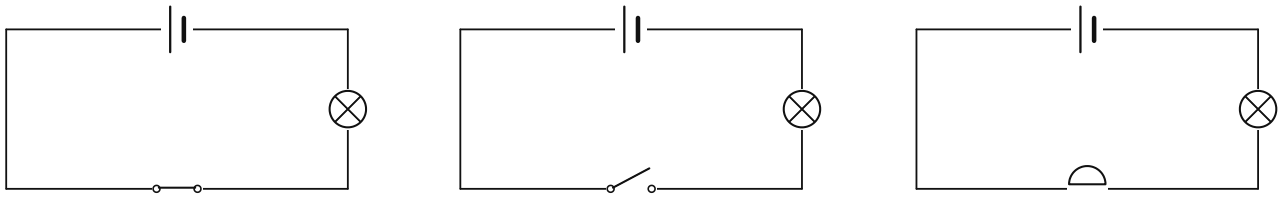
Will it work?

★☆☆
Grade 3

Learning goals

- Spot complete circuits.
- Sort magnetic metals.

1 Will the lamp light? Write yes or no. [3]



A *B* *C*

A	B	C
yes	no: switch open	yes, and the buzzer sounds

2 Sort the objects. [6]

SORT THESE

steel pin **iron nail** **aluminium foil** **plastic toy** **steel spoon** **copper wire**

Attracted by a magnet	Not attracted
iron nail	aluminium foil
steel pin	copper wire
steel spoon	plastic toy

3 Attract or repel? [3]

Poles facing	Attract or repel?
N and S	attract
N and N	repel
S and S	repel

ANSWER KEY

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LEARN · PLAY · WONDER

THINK IT THROUGH · C4 · CIRCUITS AND MAGNETS

Problem solving

★★☆
Grade 3

Learning goals

- Find faults in circuits.
- Explain with a model.

1 **Raju's torch does not work. Give three things that could be wrong.** [3]

the cell is flat; the bulb is broken; a wire is loose; the switch is broken

2 **Use the particle model to explain why the lamp goes out when you open the switch.** [2]

The gap stops the particles flowing round the loop, so no electricity reaches the lamp.

3 **Why are electric wires covered in plastic?** [2]

Plastic does not let electricity through, so it keeps us safe from shocks.

ANSWER KEY

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INVESTIGATE · C4 · CIRCUITS AND MAGNETS

Which magnet is the strongest?

★★☆
Grade 3

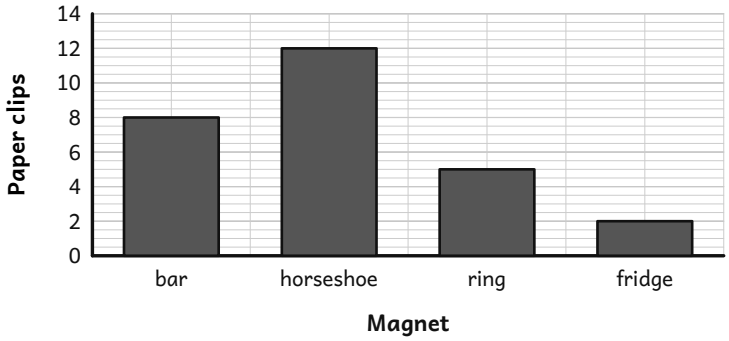
Learning goals

- Do a fair test.
- Draw a bar chart and conclude.

Class 3B hung paper clips in a chain from each magnet, one at a time, until no more would stick.

1 **Draw a bar chart.** [3]

Magnet	bar	horseshoe	ring	fridge
Paper clips	8	12	5	2



2 **Which magnet was strongest?** [1]

the horseshoe magnet

3 **Why must they use the same kind of paper clip each time?** [2]

To make it a fair test: only the magnet should change.

ANSWER KEY

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REAL-WORLD CASE STUDY · C4 · CIRCUITS AND MAGNETS

Light for every home: solar power

★★★
Grade 3

Learning goals

- See how a circuit changes lives.

Science | Technology | Community

Some villages and islands in Bangladesh are far from the electricity lines. Millions of families there use **solar home systems**: a **solar panel** on the roof, a **battery**, wires, **switches** and a few **LED lamps**.

In the day, the solar panel turns sunlight into electricity, which is stored in the battery. At night the battery lights the lamps, so children can study and families can charge their phones, without smoky kerosene lamps.

Based on solar home systems in rural Bangladesh.

1 **Name the parts of a solar home circuit.** [2]

solar panel, battery, wires, switch, lamps

2 **Why is a solar lamp better than a kerosene lamp?** [2]

no smoke, no fire danger, uses free sunlight, brighter

3 **Draw the circuit: battery, switch and lamp.** [2]

Solar lamp circuit

a loop: battery → switch → lamp → back to the battery.

ANSWER KEY

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REVIEW AND CHALLENGE · C4 · CIRCUITS AND MAGNETS

Show what you know

★★★
Grade 3

Learning goals

- Use circuit and magnet words.

1 Find the words.

L	C	I	L	A	M	P	W	E
I	L	S	M	A	C	J	O	P
D	J	U	A	I	E	I	B	I
S	A	B	G	R	C	E	L	L
W	P	D	N	O	K	B	P	P
I	W	O	E	N	G	C	O	A
T	O	U	T	K	O	H	L	H
C	B	C	L	W	I	R	E	V
H	Y	N	H	C	C	I	K	P

cell iron lamp magnet pole switch wire

2 A lamp will light if the circuit is... [1]

○ broken
● **complete**
○ wet

3 Two north poles will... [1]

○ attract
● **repel**
○ melt

4 Challenge. How could you use a magnet to separate iron nails from sand? [2]

Move the magnet over the mixture; the nails stick to it and the sand stays behind.